Kuwait University

Computer Engineering Department

CpE 420: Intro to Data Mining

Instructor: Dr. Ahmed Nasrallah
Office Hours: 12:30 – 02:00 PM, Monday, Wednesday, or by appointment

Office: ENP-S, 2nd floor, room 196

Lecture Time: 11:00 – 12:15 PM
Lecture Location: ENP-S 02 B1 041

Teaching Assistant: Eng. Anwar Al-Ebraheem
Location: TBD
Office Hours: TBD

Prerequisites: CpE 300; CpE 341

Main Textbook

Introduction to Data Mining 2nd Edition, Pang-Ning Tan, Michael Steinbach, and Vipin Kumar; Pearson, 2018. ISBN-13: 978-0133128901

Supplemental Textbook

Data Mining: Concepts and Techniques 4th Edition, Han, Jiawei, Micheline Kamber, and Jian Pei; Morgan Kaufmann, 2022. ISBN-13: 978-0128117606

Mining of Massive Datasets 2nd Edition, Jure Leskovec, Anand Rajaraman, Jeffrey David Ullman; Cambridge University Press, 2014, ISBN-13 978-1107077232

Course Description

This course is concerned with data mining techniques: classification (including decision tree induction, Bayes classification, k-nearest neighbor classifiers, linear classifiers, feature selection, Bayesian belief networks, support vector machines, rule-based and pattern-based classification), clustering (K-Means, Hierarchical, DBSCAN, Graph/Density-Based clustering), association (mining frequent patterns, correlation), preprocessing; performance evaluation; information assurance, deep learning, security and privacy issues, and other applications. Students are recommended to have a background in database management systems and statistics to be successful in this course.

Course Objectives

By the end of the course, students should be able to:

1. Demonstrate proficiency in various data mining techniques, including classification, clustering, and association.
2. Apply preprocessing methods to clean and prepare data for effective mining.
3. Evaluate the performance of data mining models using relevant metrics and techniques.
4. Understand the importance of information assurance in the context of data mining.
5. Explore web mining and its applications.
6. Analyze security and privacy issues related to data mining and propose strategies for addressing them.
7. Apply data mining techniques to real-world applications and scenarios.

Topics to be covered

Credits: 3

Total hours: 41 + (2 hours for in-class examination)

Topic No.	Topic Title	No. of hours
1	Introduction to Data Mining	3
2	Data, Measurements, and Preprocessing	6
3	Data Mining Techniques: Classification	6
4	Data Mining Techniques: Association	6
5	Data Mining Techniques: Clustering	6
6	Anomaly Detection	3
7	Deep Learning (tentative)	3
8	Security and Privacy Issues (tentative)	4
9	Applications of Data Mining (tentative)	3

Grading and Course Requirements

Assignments	20%
Quizzes	20%
Mid Term Exam	20%
Project	20%
Final Exam	20%

Grade Distribution

Grade	Range
A	≥ 93
A-	≥ 90 and < 93
B+	≥ 87 and < 90
B	≥ 83 and < 87
B-	≥ 80 and < 83
C+	≥ 77 and < 80
C	≥ 73 and < 77
C-	≥ 70 and < 73
D+	≥ 65 and < 70
D	≥ 60 and < 65
F	< 60

Important Dates

Date	Event
Thur. 07/05/2026	Last Day of Classes
Wed. 20/05/2026 12:00 to 14:00	Final Exam

Tentative Course Schedule

Week	Topic	Reading	Assignment
1 2/1-5	Course Overview & Introduction to Data Mining	Tan Ch. 1	
2 1/8-12	Data Types, Data Quality, Measurements	Tan Ch. 2	HW1 Out
3 1/15-19	Data Preprocessing: Cleaning, Integration, Transformation	Tan Ch. 3	HW1 Due
4 1/22-26	Classification I: Decision Trees, Model Evaluation	Tan Ch. 4	HW2 Out / Quiz 1
5 3/1-5	Classification II: k-NN, Naïve Bayes, Linear Models	Tan Ch. 4	HW2 Due
6 3/8-12	Classification III: SVMs, Ensemble Methods	Tan Ch. 4 / Han Ch. 8	Quiz 2
7 3/15-19	Clustering I: k-Means, Hierarchical Clustering	Tan Ch. 6	HW3 Out
8 3/22-26	Clustering II: DBSCAN, Graph-Based Clustering	Tan Ch. 7	HW3 Due / Quiz 3
9 3/29-4/2	**Midterm Exam**	Tan Ch. 7	Midterm
10 4/5-9	Association Rule Mining: Apriori, FP-Growth	Tan Ch. 7	HW4 Out
11 4/12-16	Anomaly & Outlier Detection	Tan Ch. 10	HW3 Due / Quiz 4
12 4/19-23	Deep Learning for Data Mining (Intro, NN Basics)	Han Ch. 13	Project
13 4/26-30	Security, Privacy & Ethical Issues in Data Mining	Han Ch. 11	Project Progress
14 5/3-7	Applications of Data Mining & Course Review	Case Studies	Project Demo / Review

Course Policies and Requirements

Homework Rules

- Late homework will be deducted 10% for each day it was late. (Meaning if it were late 1 minute, 10% of the final grade would be deducted)
- Soft copies of the homework are required to be submitted through Moodle. No other form of submission will be accepted.
- The submitted file needs to follow the following format {HW#}_{student_id}_{student_name}.zip, such that for student Ahmad Ibrahim it will be HW1_1234567_Ahmad_ibrahim.zip
- You must follow the rules of code indentation and add meaningful comments to the code to make it readable and understandable.
- After each homework, a student may be appointed to solve part of it on the board randomly.
- You may discuss homework problems with your classmates, but you must specify which classmate you worked with when submitting your solution. Failure to do will result in zero for both students.
- Discussing the homework does not include using someone else's code.

Exam Rules

- You must bring and display a valid Kuwait University student ID to the exams
- You are not allowed to ask questions about the exam content during the examination unless you feel a typographical error. Reading and understanding what the question is asking is part of the exam. If you need to make assumptions, then make sure you write those assumptions in your response.
- If you arrive late for an examination, you will be given the remaining amount of time to complete your exam. However, late students cannot start the examination after the first student hands in their exam. These late students will receive a FA grade in the course.

Email Policy

- Always use a proper subject title and include "CpE201: …."
- Always include your name and ID number
- Always introduce yourself before asking your questions

MS Teams Chat Policy

- Only send short one-line questions in chat. Generally, I will answer within 48 hours of receiving a message.
- If your message is too long, I will probably not read it, and you should send it by email.
- Always introduce yourself and which course you are asking about before asking any questions

AI Usage Policy
AI tools are an important part of modern computer engineering practice, and this course seeks to promote **responsible and transparent AI use**.
By default, AI tools may **not** be used for coursework. Exceptions will be clearly stated by the instructor. When permitted, students must explicitly document the **extent and nature of AI assistance**. Students are expected to critically evaluate AI-generated content and demonstrate independent understanding.

When AI use **is explicitly permitted**, students **must clearly disclose**:

- **Which AI tool(s)** were used
- **How the tool was used** (e.g., brainstorming, code generation, debugging, explanation)
- **An estimate of the proportion of the submitted work influenced by AI** (e.g., "~20% AI-assisted")

Attendance
Attendance in this class is required. While no specific grade is given for attendance, absences and late attendance will negatively affect your in-class experience. Your responsibility is to seek out help from classmates to fill you in on missed materials.
In addition to my classroom policies, you are also expected to abide by the guidelines of the University. Every student in this course must abide by the Kuwait University Policy on Attendance (published in the Student Guide). A copy of the student guide can be accessed online on:
http://vpaa.ku.edu.kw/ar/documents/KU%20ByLaws/Students/Curriculum_Regulations.pdf

All enrolled students are expected to attend the class. According to Kuwait University rules, **a student who misses 5 or more classes will automatically get an FA grade** and will not be allowed to attend the final exam.

Academic Integrity

The University's code of academic integrity is designed to ensure that the principles of academic honesty and integrity are upheld. All students are expected to adhere to this code. All acts of academic dishonesty will be dealt with in accordance with the provisions of this code. Every student in this course must abide by the Kuwait University Policy on Cheating and Plagiarism (published in the Student Guide). A copy of the student guide can be accessed online on:
http://vpaa.ku.edu.kw/ar/documents/KU%20ByLaws/Students/Curriculum_Regulations.pdf
In particular,

- You should neither give nor receive assistance from anyone in taking the quizzes, assignments, projects, and final exam.
- You should immediately report to me any act of academic dishonesty that you may observe. Your anonymity will be protected.

Cheating includes (but is not limited to):

- Using unauthorized notes during the examination.
- Using unauthorized devices during the exam.
- Communicating with anyone besides the instructor or exam proctor during an exam.
- Looking at another student's work during the exam.
- Copying or submitting another student's work or from an AI tool.
- Submitting someone else's solutions as your own.
- Acquiring help in the project from anyone other than the instructor or teacher's assistant.
- Have someone else take the exam on your behalf.

I will not tolerate any attempt at cheating, and they will be directly reported.

Other Course Policies

Your fellow students and I need 100% of your attention and mindshare while the class is in session. Toward this end:

- For each class meeting, please arrive sufficiently ahead of the official start time to collect any handouts or previous quizzes that have been graded and get yourself ready with your notes and papers.
- Please do not walk around or out of the classroom while class is in session.
- Turn off phone ringtones, including vibration during class.

Note: Please DO NOT send anyone (a family member or any other person) to talk to me about giving you a higher grade. Any student who violates this rule will not receive class bonuses (if any).

Note: This syllabus is subject to change without notice.

Special Needs: If you have a disability, special needs, or require a specific grade, you should bring this to my attention as soon as possible, but not later than the second week of class. This includes getting a particular grade in this class.